

Interaction-driven altermagnetism in the checkerboard Hubbard model

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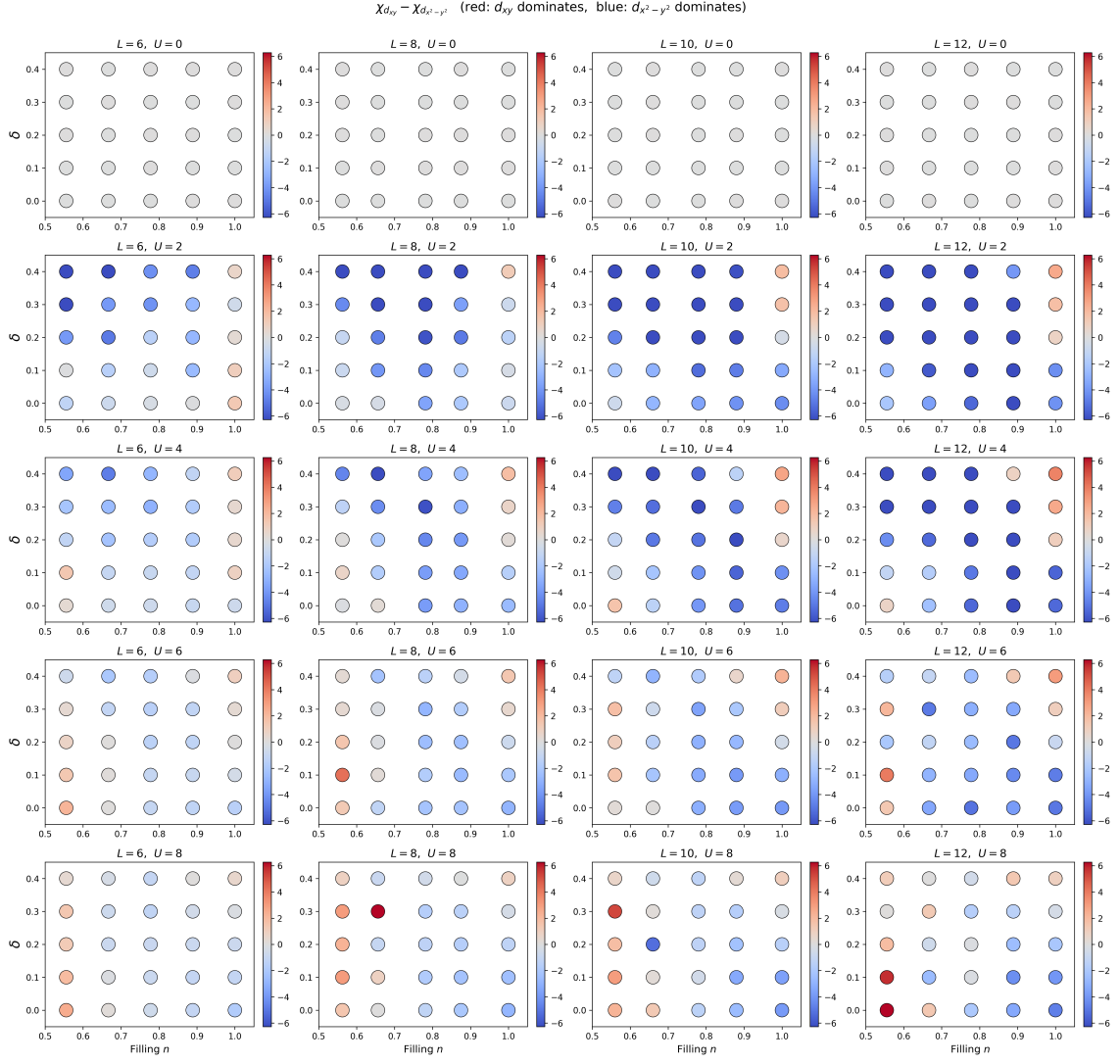
Central claim

An altermagnet suppresses the superconducting pairing channel that shares its own symmetry. On the checkerboard lattice the magnetic form factor is fixed by geometry to be d_{xy} , and it is the d_{xy} pairing channel that is destroyed, while the orthogonal

D. Pairing tracks the order parameter. $\chi_{d_{xy}}$ against m_{AM} at fixed filling. This is what turns a dependence on δ into a statement about the magnetic order itself.

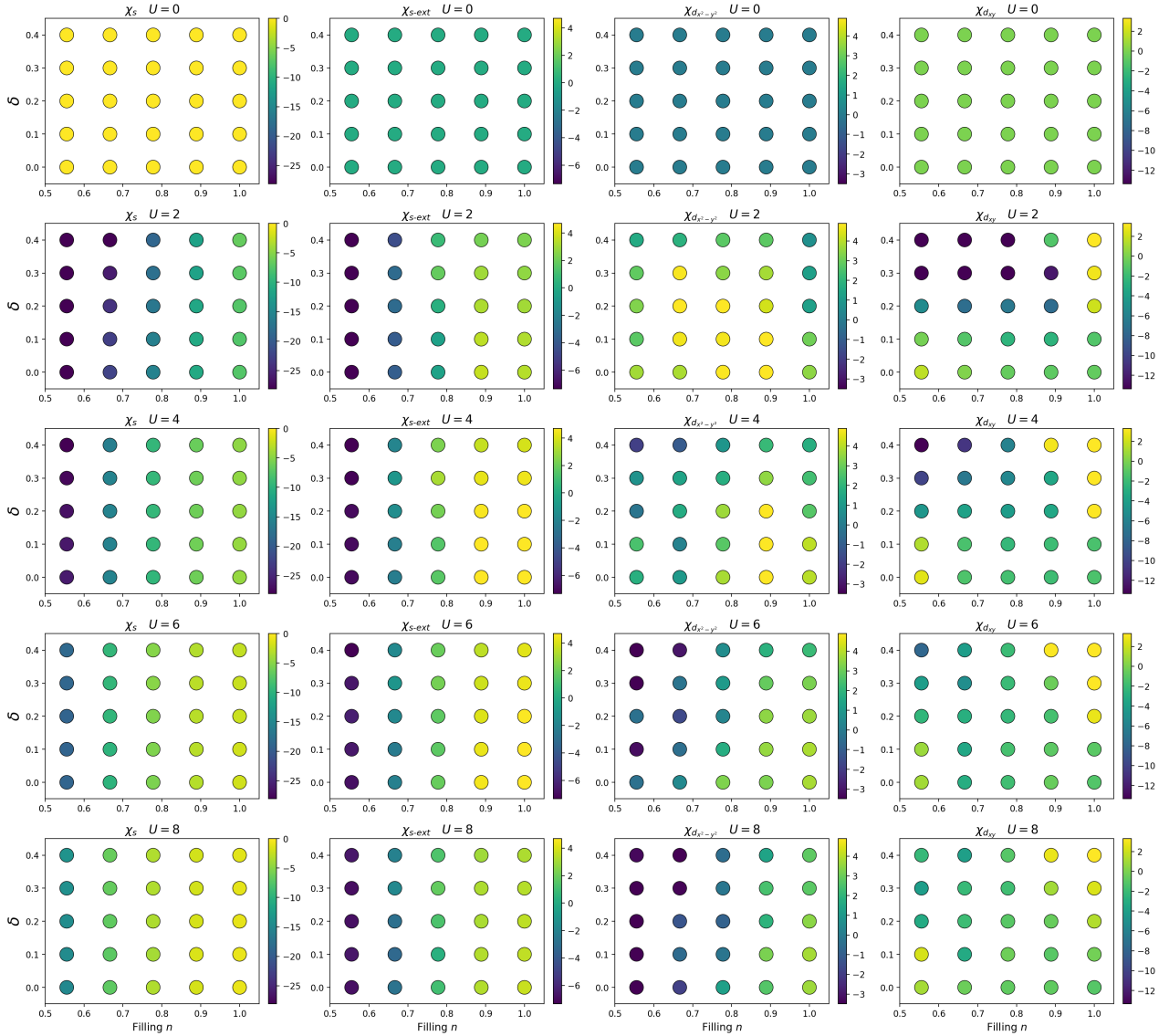
E. Half filling. The dominant channel changes near $\delta \approx 0.2$ at every U . Stated with the open-shell caveat.

F. Finite-size analysis. $L = 6$ to 12 at six seeds. The selection strengthens with system size.

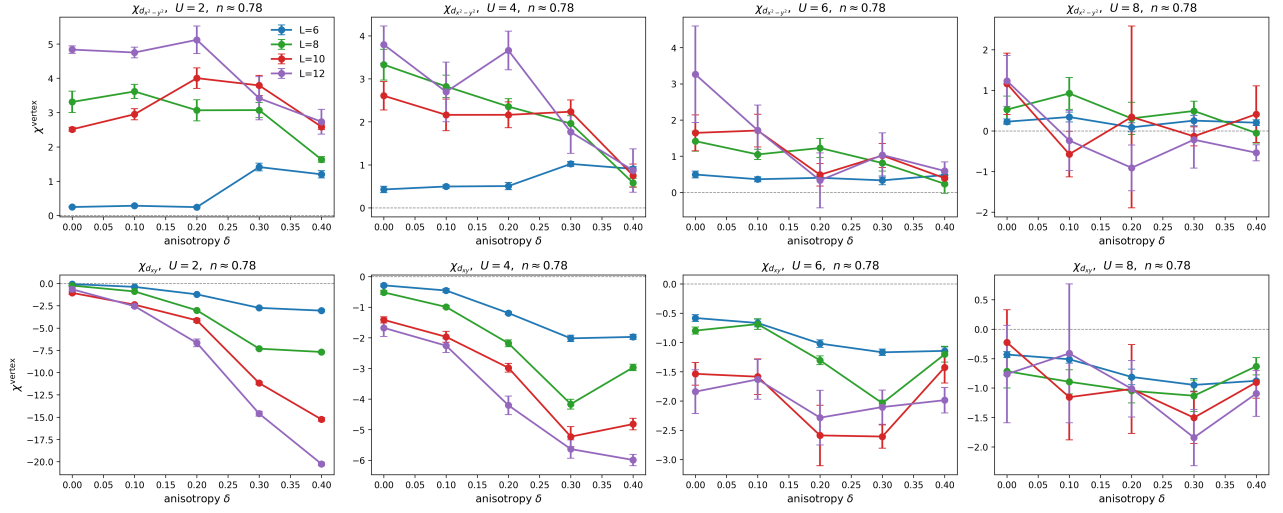


Interaction dependence and the $U = 0$ null test

The $U = 0$ row of the figure below is zero to machine precision in all four channels. It is a validation of the vertex construction rather than data, and it is the cleanest statement in the paper that the pairing is interaction driven.



Finite-size test: $L=6$ says $d_{x^2-y^2}$ is ENHANCED by δ ; $L=8,10,12$ all say SUPPRESSED



Pairing versus the altermagnetic order parameter

At fixed filling, χ_{dxy} tracks m_{AM} closely in the doped region, while $d_{x^2-y^2}$ shows no such dependence. Spearman rank correlations:

Filling	rho with d(xy)	rho with d(x ² -y ²)
n = 0.556		

Open items and decisions

Runs in progress

Run	Purpose	Status
Magnetic cube, $L = 12, 10, 8$	Sec. III.C, and confirms Sec. IV.D	$U = 0$ complete; $U = 4$ first block done; $U = 2, 6, 8$ running
Dense filling scan, $L = 12, U = 4$	Locates the boundary in <	